Lecture 3: Hyperbolic Functions

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what is the hyperbolic function.
- ☐ Find the derivative and the integral of the hyperbolic function.
- ☐ Find the inverse hyperbolic function.
- ☐ Find the derivative of the inverse hyperbolic function.
- ☐ Perform basic integration using integration formula involving inverse hyperbolic function.

Introduction

Any function $f : \mathbb{R} \to \mathbb{R}$ can be written as a sum of an even function and an odd function.

$$f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$$

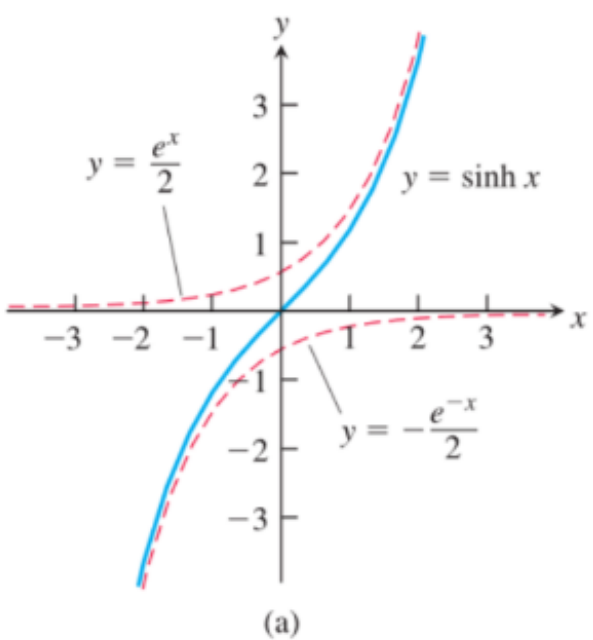
If we let $f(x) = e^x$, then we have

$$e^x = \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2}$$

Hyperbolic Functions

Definition: We define the hyperbolic sine and hyperbolic cosine functions by the odd and even parts of $f(x) = e^x$, respectively.

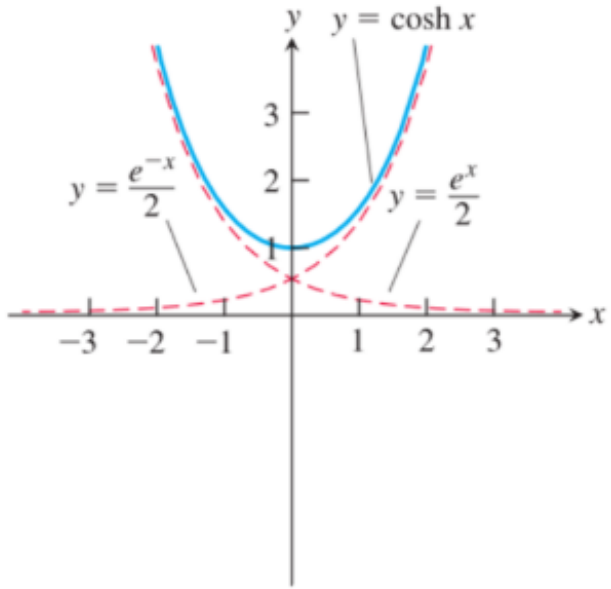
$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \text{and} \quad \cosh x = \frac{e^x + e^{-x}}{2}.$$



(a)

Hyperbolic sine:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

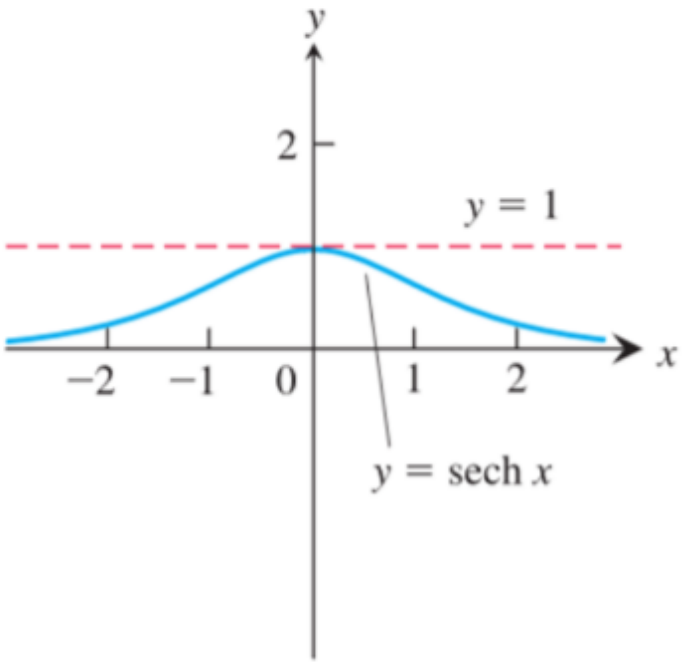


(b)

Hyperbolic cosine:

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

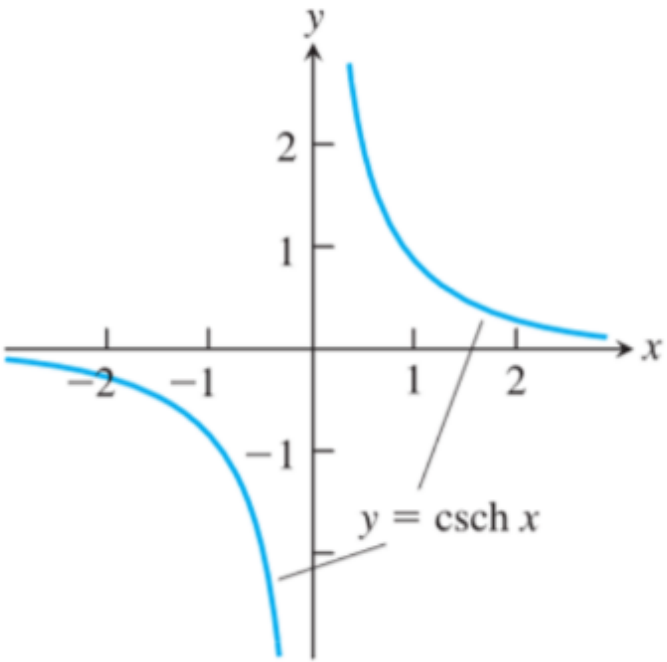
Hyperbolic Functions



(d)

Hyperbolic secant:

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

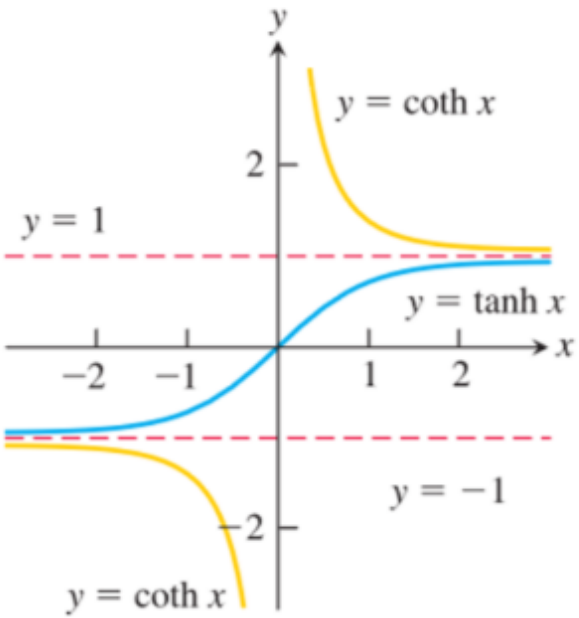


(e)

Hyperbolic cosecant:

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

Hyperbolic Functions



(c)

Hyperbolic tangent:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Hyperbolic cotangent:

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

Hyperbolic Functions

Some identities for hyperbolic functions

- $\cosh^2 x - \sinh^2 x = 1$
- $\sinh 2x = 2 \sinh x \cosh x$
- $\cosh 2x = \cosh^2 x + \sinh^2 x$
- $\cosh^2 x = \frac{\cosh 2x + 1}{2}$

Hyperbolic Functions

Some identities for hyperbolic functions

- $\sinh^2 x = \frac{\cosh 2x - 1}{2}$
- $\tanh^2 x = 1 - \operatorname{sech}^2 x$
- $\coth^2 x = 1 + \operatorname{csch}^2 x$

Derivatives of Hyperbolic Functions

1. $\frac{d}{dx}(\sinh x) = \cosh x$
2. $\frac{d}{dx}(\cosh x) = \sinh x$
3. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$
4. $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$
5. $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$
6. $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$

Derivatives of Hyperbolic Functions

Example: Evaluate

1. $\frac{d}{dx}\cosh(1+x^2)$
2. $\frac{d}{dx}(\tanh\sqrt{1+x^2})$
3. $\frac{d}{dx}(x^2\sinh x)$
4. $\frac{d}{dx}(e^x\sinh x)$
5. $\frac{d}{dx}[(x^2-1)\operatorname{csch}(\ln x)]$

Derivatives of Hyperbolic Functions

Integrals of Hyperbolic Functions

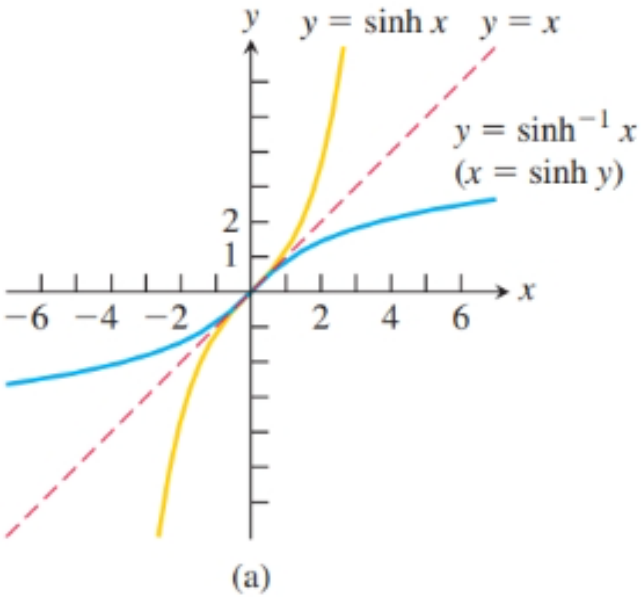
1. $\int \sinh x \, dx = \cosh x + C$
2. $\int \cosh x \, dx = \sinh x + C$
3. $\int \operatorname{sech}^2 x \, dx = \tanh x + C$
4. $\int \operatorname{csch}^2 x \, dx = -\coth x + C$
5. $\int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x + C$
6. $\int \operatorname{csch} x \coth x \, dx = -\operatorname{csch} x + C$

Integrals of Hyperbolic Functions

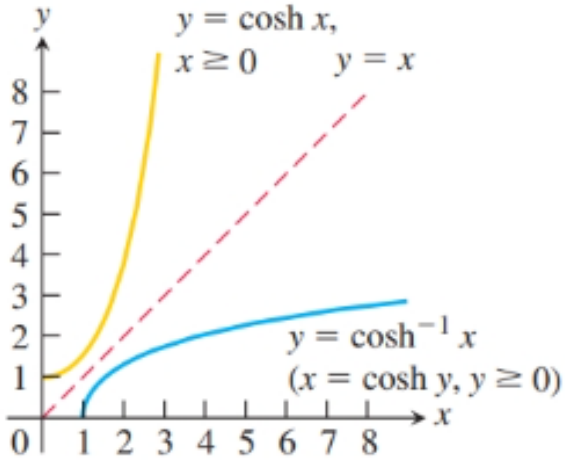
Example: Evaluate

1. $\int \coth 5x \, dx$
2. $\int_0^1 \sinh^2 x \, dx$
3. $\int_0^{\ln 2} 4e^x \sinh x \, dx$
4. $\int_0^{\frac{\ln 2}{3}} \cosh 3x \, dx$

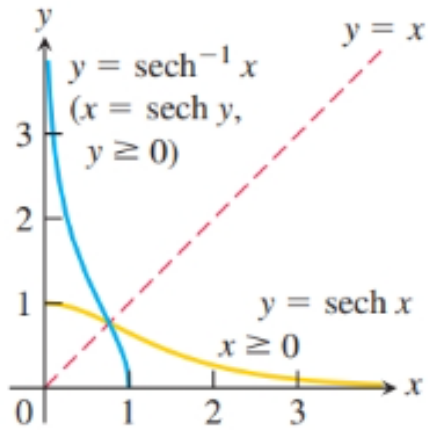
Inverse Hyperbolic Function



(a)

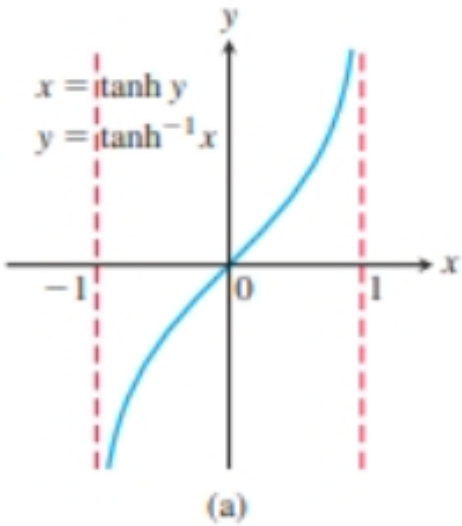


(b)

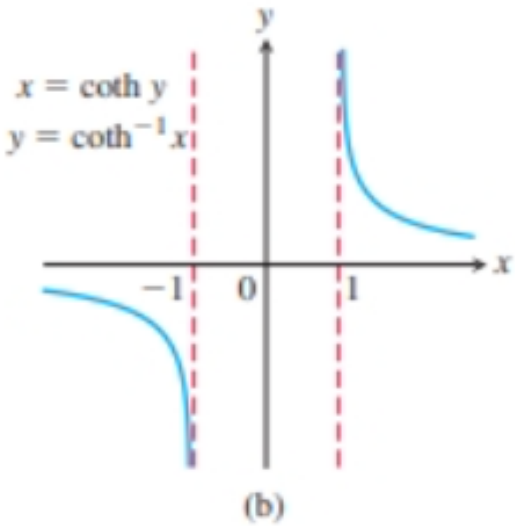


(c)

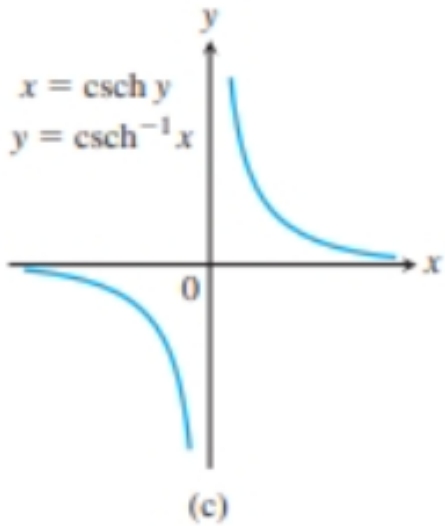
Inverse Hyperbolic Function



(a)



(b)



(c)

Inverse Hyperbolic Functions

$$\sinh(\sinh^{-1} x) = x \ \forall x \in \mathbb{R} \quad , \quad \sinh^{-1}(\sinh x) = x \ \forall x \in \mathbb{R}$$

$$\cosh(\cosh^{-1} x) = x \ \forall x \in [1, \infty) \quad , \quad \cosh^{-1}(\cosh x) = x \ \forall x \in [0.\infty)$$

$$\tanh(\tanh^{-1} x) = x \ \forall x \in [-1, 1] \quad , \quad \tanh^{-1}(\tanh x) = x \ \forall x \in \mathbb{R}$$

$$\operatorname{sech}(\operatorname{sech}^{-1} x) = x \ \forall x \in (0, 1] \quad , \quad \operatorname{sech}^{-1}(\operatorname{sech} x) = x \ \forall x \in [0, \infty)$$

$$\coth(\coth^{-1} x) = x \ \forall x \in \mathbb{R} \setminus [-1, 1] \quad , \quad \coth^{-1}(\coth x) = x \ \forall x \in \mathbb{R} \setminus \{0\}$$

$$\operatorname{csch}(\operatorname{csch}^{-1} x) = x \ \forall x \in \mathbb{R} \setminus \{0\} \quad , \quad \operatorname{csch}^{-1}(\operatorname{csch} x) = x \ \forall x \in \mathbb{R} \setminus \{0\}$$

Inverse Hyperbolic Functions

Some identities for inverse hyperbolic functions

- $\operatorname{sech}^{-1} x = \cosh^{-1}(\frac{1}{x}), x \in (0, 1]$
- $\operatorname{csch}^{-1} x = \sinh^{-1}(\frac{1}{x}), x \in (-\infty, 0) \cup (0, \infty)$
- $\coth^{-1} x = \tanh^{-1}(\frac{1}{x}), x \in (-\infty, -1) \cup (1, \infty)$

Derivatives of Inverse Hyperbolic Functions

1. $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$
2. $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$ $(x > 1)$
3. $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$ $(|x| < 1)$
4. $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$ $(|x| > 1)$
5. $\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}$ $(0 < x < 1)$
6. $\frac{d}{dx}(\operatorname{csch}^{-1} x) = -\frac{1}{|x|\sqrt{1+x^2}}$ $(x \neq 0)$

Derivative of Inverse Hyperbolic Function

Example: Evaluate

1. $\frac{d}{dx}(\sinh^{-1} x^2)$
2. $\frac{d}{dx}(\tanh^{-1}(\sin 2x))$
3. $\frac{d}{dx}(\cosh^{-1}(x^2 - 1))$
4. $\frac{d}{dx}(\coth^{-1}(3x^2))$
5. $\frac{d}{dx}(\ln(x)\cosh(x))$

Integration Formulas

Integrals evaluated with inverse trigonometric functions

1. $\int \frac{dx}{\sqrt{a^2+x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C, \qquad a > 0$

2. $\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C, \qquad x > a > 0$

3. $\int \frac{dx}{a^2-x^2} = \begin{cases} \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + C, & x^2 < a^2 \\ \frac{1}{a}\coth^{-1}\left(\frac{x}{a}\right) + C, & x^2 > a^2 \end{cases}$

4. $\int \frac{dx}{x\sqrt{a^2-x^2}} = -\frac{1}{a}\operatorname{sech}^{-1}\left(\frac{x}{a}\right) + C, \qquad 0 < x < a$

5. $\int \frac{dx}{x\sqrt{a^2+x^2}} = -\frac{1}{a}\operatorname{csch}^{-1}\left|\frac{x}{a}\right| + C, \qquad x \neq 0 \text{ and } a > 0$

Integration Formulas

Example: Evaluate

1. $\displaystyle\int \frac{2\,dx}{\sqrt{3+4x^2}}$
2. $\displaystyle\int \frac{dx}{\sqrt{x^2+4}}$
3. $\displaystyle\int \frac{dx}{\sqrt{9x^2+4}}$
4. $\displaystyle\int_0^{\frac{1}{2}} \frac{dx}{1-x^2}$